Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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Research Interests

My research sits at the intersection of empirical software engineering, code comprehension, and human–AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

Education

Oregon State University — Corvallis, OR, United States
Ph.D. in Computer Science, School of EECS — *Sep 2023 – Jun 2027 (expected)*
Advisor: Dr. Christopher Hundhausen at the SEAL Lab
Research Areas: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction

Arizona State University — Tempe, AZ, United States
M.S. in Software Engineering — *Aug 2021 – May 2023*
Advisor: Dr. Robert LiKamWa
Thesis: Networked System for Volumetric Athletic Coaching in Augmented Reality

Guangxi University of Science and Technology — Liuzhou, China
B.E. in Software Engineering — *Sep 2016 – Jun 2020*

Southern Cross University — Gold Coast, Australia
B.E. in Information Technology — *Sep 2017 – Jun 2020*

Publications

Journal Articles & Conference Papers

Yunhan Qiao, Md Istiak Hossain Shihab, and Christopher Hundhausen. **A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice.** ***TOCE****: ACM Transactions on Computing Education*, 26(2), Article 24, 1–33, 2026. [PDF] [DOI]
Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.

Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, Sanjay Chandrasekaran, **Yunhan Qiao**, and Summit Haque. **Using SlackBot-Prompted Reflections to Promote Contribution Equality in Team Software Development Projects.** ***TOCE****: ACM Transactions on Computing Education*, 2026. [PDF] [DOI]

Md Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, **Yunhan Qiao**, and Brian Wise Mulanda. **The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks.** ***ICER'25****: Proceedings of the 2025 ACM Conference on International Computing Education Research*, Vol. 1, 407–420, Aug. 2025. [PDF] [DOI]

Manish Motwani, Aakash Kulkarni, **Yunhan Qiao**, Matthew Davis, and Ziyan Chen. **LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.** ***AIxSE'25****: IEEE International Conference on AI x Software Engineering*, 59–66, 2025. [PDF] [DOI]

Under Review

Yunhan Qiao, Md Istiak Hossain Shihab, Summit Haque, and Christopher Hundhausen. **Faster but Not Wiser: GitHub Copilot Decouples Programming Performance from Code Comprehension in Brownfield Tasks.** ***TOCE****: ACM Transactions on Computing Education*, under review. [arXiv]

Yunhan Qiao, Summit Haque, and Christopher Hundhausen. **Beyond the Prompt: Linking What Developers Ask, Do, and Understand with Coding Agents.** ***SANER'27****: IEEE International Conference on Software Analysis, Evolution and Reengineering*, under review. [arXiv]

Under Submission

O. Schutz, **Yunhan Qiao**, Fatima A. Moussaoui, Rudrajit Choudhuri, and Anita Sarma. **Heart and Mind in the Loop: Why We Vibe When We Work with AI.** ***CHASE'27****: International Conference on Cooperative and Human Aspects of Software Engineering*, under submission.

Teaching Experience

Graduate Teaching Assistant, School of EECS — Oregon State University
Undergraduate software engineering courses — *2024 – current*
Courses: CS 362 Software Engineering II; CS 461 Senior Software Engineering Project; CS 467 Online Capstone Project; CS 492 Mobile Software Development

Professional Experience

Software Engineering Intern, Mocha Software Nanjing, China
Software development *Aug 2020 – Nov 2020*

Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

Awards & Scholarships

Oregon State University Scholarly Presentation Award *2026*
Oregon State University EECS Graduate Scholarship *2023 – current*
Guangxi University of Science and Technology Merit Award *2018*

Academic Service

SIGCSE TS 2027, Main Track, Program Committee

SANER 2027, Short Papers and Posters Track, Program Committee

ACM Transactions on Computing Education (TOCE), Reviewer

Skills & Coursework

Research Methods: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
Programming: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
Frameworks & Tools: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
Coursework: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

Languages

English (Fluent), Mandarin Chinese (Native)